

Prac. Problem for Quiz 1

1. $n = \frac{Z_{\alpha/2}^2}{4e^2}$ $\alpha = 0.05$
 $= \frac{(1.96)^2}{4(0.04)^2}$ $\alpha/2 = 0.025$
 $= \frac{3.8416}{0.0064} = 600.25 \approx 601$ $Z_{0.025} = 1.96$
 (if p was unknown)

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$e = Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$0.04 = 1.96 \sqrt{\frac{0.2275}{n}}$$

$$\frac{0.04}{1.96} = \sqrt{\frac{0.2275}{n}}$$

$$n = \frac{0.2275}{0.0004} = 568.75 \approx 569$$

for better
approx.
first.
calc. the
formula.

$n = ?$
 $\hat{p} = 0.35$
 $(1-\hat{p}) = 0.65$
 $\sqrt{\hat{p}(1-\hat{p})} = 0.2275$
 $e = 0.04$
 $\sqrt{Z_{0.025}} = 1.96$

$$\Rightarrow (0.0204)^2 = \frac{0.2275}{n}$$

2. $e = 0.03$ $\alpha = 0.01$, $\alpha/2 = 0.005$, $Z_{0.005} = 2.576$

$$n = \frac{Z_{\alpha/2}^2}{4e^2} = \frac{(2.576)^2}{4(0.03)^2} = \frac{6.635776}{0.0036} = 1843.27111 \approx 1844$$

3. $X_1, \dots, X_n$, $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$

$$\hat{\mu} = \frac{X_1 + X_2 + X_3}{3}$$

$$\text{Bias}(\hat{\mu}) = ?$$

$$\text{MSE}(\hat{\mu}) = ?$$

$$\text{Bias}(\hat{\mu}) = E[\hat{\mu}] - \mu$$

$$\text{Bias}(\hat{\mu}) = E[\hat{\mu}] - \mu = E\left[\frac{X_1 + X_2 + X_3}{3}\right] - \mu$$

$$= \frac{1}{3} \times 3\mu - \mu = 0$$

$$\text{Bias}(\hat{\mu}) = 0$$

$$\text{MSE}(\hat{\mu}) = \underbrace{(\text{Bias}(\hat{\mu}))^2}_0 + \text{Var}(\hat{\mu})$$

$$\text{MSE}(\hat{\mu}) = \text{Var}(\hat{\mu}) \Rightarrow \text{Var}\left(\frac{X_1 + X_2 + X_3}{3}\right)$$

$$\frac{1}{9} \text{Var}(X_1 + X_2 + X_3) = \frac{1}{9} \times 9\sigma^2 = \sigma^2$$

4. X & Y are indep. RV. $E[X] = 30$, $\text{Var}(X) = 4$
 $E[Y] = 0$, $\text{Var}(Y) = 9$

$$\hat{\theta} = \frac{X + 2Y}{5}$$

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - 0$$

$$= 0 - 0$$

$$\text{Bias}(\hat{\theta}) = 0$$

$$E[\hat{\theta}] = E\left[\frac{X + 2Y}{5}\right]$$

$$= \frac{1}{5} E[X + 2Y]$$

$$= \frac{1}{5} (E[X] + 2E[Y])$$

$$\text{Var}(\hat{\theta}) = \text{Var}\left(\frac{X + 2Y}{5}\right) = \frac{1}{5} \times 5 = 1$$

$$= \frac{1}{25} [\text{Var}(X + 2Y)] = \frac{1}{25} (\text{Var}(X) + 4\text{Var}(Y))$$

$$= \frac{4 + 36}{25} = \frac{40}{25} = \frac{8}{5} = 1.6$$

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2 = 1.6$$

5. X & Y are 2 indep. RV. $E[X] = 20$, $\text{Var}(X) = 5$
 $E[Y] = 40$, $\text{Var}(Y) = 8$

$$\hat{\theta} = \frac{3X + Y}{10}$$

$$E[\hat{\theta}] = E\left[\frac{3X + Y}{10}\right] = \frac{1}{10} (3E[X] + E[Y])$$

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - 0$$

$$= \frac{120}{10} = 12$$

$$= 12 - 0 = 12, \text{ Yes, unbiased.}$$

$$\begin{aligned} \text{Var}(\hat{\theta}) &= \text{Var}\left(\frac{3X+Y}{10}\right) = \frac{1}{100} (\text{Var}(3X+Y)) \\ &= \frac{9 \text{Var}(X) + \text{Var}(Y)}{100} = \frac{45+8}{100} = \frac{53}{100} \\ &= 0.53 \\ \text{MSE}(\hat{\theta}) &= 0.53 \end{aligned}$$

divide by 100

6. $E[\hat{\theta}_1] = 0, E[\hat{\theta}_2] = 0+1$
 $\text{Var}(\hat{\theta}_1) = 4, \text{Var}(\hat{\theta}_2) = 1$

$$\text{Bias}(\hat{\theta}_1) = E[\hat{\theta}_1] - 0 = 0 - 0 = 0$$

$$\text{Var}(\hat{\theta}_1) = 4$$

$$\begin{aligned} \text{MSE}(\hat{\theta}_1) &= \text{Var}(\hat{\theta}_1) + (\text{Bias}(\hat{\theta}_1))^2 \\ &= 4 + 0 = 4 \end{aligned}$$

$$\text{Bias}(\hat{\theta}_2) = E[\hat{\theta}_2] - 0 = 0+1 - 0 = 1$$

$$\text{Var}(\hat{\theta}_2) = 1$$

$$\text{MSE}(\hat{\theta}_2) = 1 + (1)^2 = 2$$

$\hat{\theta}_2$ is preferred.

7. $E[\hat{\theta}] = 0+2, \text{Var}(\hat{\theta}) = 3$

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - 0 = 0+2 - 0 = 2$$

$$\text{Var}(\hat{\theta}) = 3$$

$$\begin{aligned} \text{MSE}(\hat{\theta}) &= \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2 \\ &= 3 + 4 = 7 \end{aligned}$$

if $\text{Bias}(\hat{\theta}) = 0$, then $\text{MSE}(\hat{\theta}) = 3$

8. $n=25 (\leq 30)$ population is normal
 not normal

$$\bar{x} = 48, S = 5, \alpha = 0.05$$

$$\alpha/2 = 0.025$$

$$t_{0.025, 24} = 2.064$$

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$$

$$\sqrt{5} = 5$$

$$48 \pm \frac{(2.064) \cancel{5}}{5} \Rightarrow 48 \pm 2.064$$

$$\Rightarrow (45.936, 50.064)$$

9. $n=36$, $\bar{X}=72$, $S=12$, $\alpha=0.1$, $\alpha/2=0.05$
 $t_{0.05, 35} = 1.690$

$$\sqrt{n} = \sqrt{36} = 6$$

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} \quad \frac{S}{\sqrt{n}} = \frac{12}{6} = 2$$

$$72 \pm (1.690) 2 = 72 \pm 3.38 = (68.62, 75.38)$$

10. $n=16$, $\bar{X}=10.4$, $S^2=0.25$, $\alpha=0.05$

this is mistake

$$\cancel{S^2=0.0625}$$

$$\alpha/2 = 0.025$$

$$1-\alpha/2 = 0.975$$

$$n-1 = 15$$

$$\frac{(n-1)S^2}{\chi^2_{\alpha/2, n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\alpha/2, n-1}}$$

$$(n-1)S^2 = 15 \times (0.25) = \cancel{0.9375} = 3.75$$

$$\chi^2_{0.025, 15} = 6.262$$

$$\chi^2_{0.975, 15} = 27.488$$

$$\left\{ \begin{array}{l} \frac{0.9375}{6.262} < \sigma^2 < \frac{0.9375}{27.488} \\ 0.14971 < \sigma^2 < 0.03410 \end{array} \right\} \text{ not correct because my } S^2 \text{ does not lie in this range.}$$

$$\frac{0.9375}{27.488} < \sigma^2 < \frac{0.9375}{6.262}$$

$$0.03410 < \sigma^2 < 0.14971$$

$$0.13 < \sigma^2 < 0.59$$

11. $n_1 = 12$, $\bar{x}_1 = 25$, $s_1^2 = 4$
 $n_2 = 10$, $\bar{x}_2 = 22$, $s_2^2 = 5$

$$\alpha = 0.05, \alpha/2 = 0.025$$

var. is equal $\Rightarrow t$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(11)(4) + (9)(5)}{20} = \frac{44 + 45}{20}$$

$$s_p^2 = \frac{89}{20} \Rightarrow s_p = \sqrt{\frac{89}{20}} = 2.109502$$

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, n_1 + n_2 - 2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$t_{0.025, 20} = 2.086 \cdot \sqrt{\frac{1}{12} + \frac{1}{10}} = \frac{10 + 12}{120}$$

$$\sqrt{\frac{22}{120}} = 0.42817$$

$$3 \pm (2.086)(2.109502)(0.42817)$$

$$3 \pm 1.88413 \Rightarrow (1.11587, 4.88413)$$

12. $n_1 = 15$, $s_1^2 = 6$, $n_2 = 12$, $s_2^2 = 3$

$$\alpha = 0.1, \alpha/2 = 0.05$$

$$v_1 = 14$$

$$v_2 = 11$$

$$\frac{s_1^2}{s_2^2} \frac{1}{F_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} F_{\alpha/2}(v_2, v_1)$$

$$\frac{s_1^2}{s_2^2} = \frac{6}{3} = 2$$

$$F_{0.05}(14, 11) = 2.72$$

$$F_{0.05}(11, 14) = 2.6$$

$$\frac{2}{2.72} < \frac{\sigma_1^2}{\sigma_2^2} < 2(2.6)$$

$$0.7352 < \frac{\sigma_1^2}{\sigma_2^2} < 5.2$$

13. $n_1 = 20, \bar{x}_1 = 8.2, s_1^2 = 1.6, \alpha = 0.1$
 $n_2 = 18, \bar{x}_2 = 7.5, s_2^2 = 1.4, \alpha/2 = 0.05$

2 popⁿ are normal with equal var $\Rightarrow t, s_p$

$\bar{x}_1 - \bar{x}_2 = 8.2 - 7.5 = 0.7$
 $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2} = \frac{(20)(1.6) + (18)(1.4)}{36}$
 $= \frac{32 + 25.2}{36} = 1.5888$
 $s_p = 1.26051$
 $t_{0.05, 36} = 0.32489$

$0.7 \pm \left(t_{0.05, 36} \right) (1.26051) (0.39489)$
 1.688

$\Rightarrow 0.7 \pm 0.82981 \Rightarrow (+0.02981, 1.52981)$

$\mu_1 - \mu_2 > 0$, first battery will have higher life.

14. $n = 10, v = 9, s^2 = 16, \alpha = 0.05, \alpha/2 = 0.025$

$\frac{(n-1)s^2}{\chi^2_{1-\alpha/2, n-1}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{\alpha/2, n-1}}$

$\chi^2_{0.975, 9} = 19.023, \chi^2_{0.025, 9} = 2.7$

$9 \times 16 = 144$

$\frac{144}{19.023} < \sigma^2 < \frac{144}{2.7}$

$7.5697 < \sigma^2 < 53.33$

15. $X_1, \dots, X_n \sim \text{Exp}(\lambda)$

$f(x) = \frac{1}{\lambda} e^{-x/\lambda}$

$$E[X] = 0 = \lambda \quad f(x) = \lambda e^{-\lambda x}$$

$$E[X] = 0 \Rightarrow \bar{X} = 0 \quad \therefore, \hat{\theta}_{\text{MOM}} = \bar{X}$$

$$E[\bar{X}] = E[X] = 0, \quad \text{bias} = 0$$

$\therefore$, therefore, unbiased

16. $X_1 \dots X_n \sim \text{Geometric}$

$$P(X=x) = p(1-p)^{x-1} \quad x = 0, 1, 2$$

$$E[X] = \frac{1}{p} \quad \bar{X} = \frac{1}{p} \rightarrow p = \frac{1}{\bar{X}}$$

$$\therefore \hat{p}_{\text{MOM}} = \frac{1}{\bar{X}}$$

17. $X_1 \dots X_n \sim N(\mu, \sigma^2) \quad \sigma^2 \Rightarrow \text{constant}$

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$L(\mu) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right)$$

$$l(\mu) = C - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

$$\frac{dl}{d\mu} = \frac{1}{\sigma^2} \sum (x_i - \mu) \Rightarrow \sum x_i - n\mu = 0$$

$$\mu = \frac{1}{n} \sum x_i \Rightarrow \hat{\mu}_{\text{MLE}} = \bar{x}$$

18. $X_1 \dots X_n \sim \text{Exp}(\lambda)$

$$L(\theta) = \prod_{i=1}^n \theta e^{-\theta x_i}$$

$$f(x) = \theta e^{-\theta x}$$

$$L(\theta) = \theta^n e^{-\theta \sum x_i} \Rightarrow \ell(\theta) = n \log \theta - \theta \sum x_i$$

$$\ell'(\theta) = \frac{n}{\theta} - \sum x_i \Rightarrow 0 \Rightarrow \frac{n}{\theta} = \sum x_i$$

$$\hat{\theta} = \frac{n}{\sum x_i} = \frac{1}{\bar{X}} \Rightarrow \hat{\theta}_{\text{MLE}} = \frac{1}{\bar{X}}$$

19. $X_1 \dots X_n \sim U(0, \theta)$

(a) $E[X] = \frac{\theta}{2}, \quad \bar{X} = \frac{\theta}{2} \Rightarrow \hat{\theta}_{\text{MOM}} = 2\bar{X}$

$$(b) \quad f(x) = \frac{1}{\theta} \quad 0 < x < \theta$$

$$L(\theta) = \theta^{-n} I(\theta \geq X_n)$$

$$X_n = \max(X_1, \dots, X_n)$$

$$\hat{\theta}_{MLE} = X_n$$

20. (a) $X_1, \dots, X_n \sim U[-\theta, \theta]$

$$E[X] = 0, \quad \bar{X} = 0$$

That means first moment can't produce a MOM estimator of θ .

$$(b) \quad -\theta < X_i < \theta \Rightarrow \theta \geq |X_i|$$

$$\hat{\theta}_{MLE} = \max_{1 \leq i \leq n} |X_i|$$

21. $X_1, \dots, X_n \sim U[0, \theta+1]$

$$f(x) = 1, \text{ when } 0 < x < \theta+1$$

$$\hat{\theta}_{MLE} \in [X_{n-1}, X_n]$$

22. $X_1, \dots, X_n \sim U[0, 2\theta]$

$$(a) \quad E[X] = \frac{2\theta}{2} = \theta, \quad \bar{X} = \theta$$

$$\hat{\theta}_{MOM} = \bar{X}$$

$$(b) \quad f(x) = \frac{1}{2\theta}$$

$$X_n \leq 2\theta$$

$$\theta \geq \frac{X_n}{2}$$

$$L(\theta) \propto \theta^{-n}$$

$$\hat{\theta}_{MLE} = \frac{X_{(n)}}{2}$$

23. $X_1, \dots, X_n \sim U(\theta-1, \theta+1)$

$$\theta-1 < X_i < \theta+1$$

$$X_i > \theta-1 \Rightarrow \theta < X_i+1$$

$$X_i < \theta+1 \Rightarrow \theta > X_i-1$$

$$X_{n-1} \leq \theta \leq X_1+1$$

$$\hat{\theta}_{MLE} \in [X_{n-1}, X_1+1]$$

24. $X_1, \dots, X_n \quad f(x) = \frac{2x}{\theta^2}$

(a) $E[X] = \frac{2\theta}{3}$, $\bar{X} = \frac{2\theta}{3} \Rightarrow \theta = \frac{3\bar{X}}{2}$ (MoM)

(b) $f(x) = \frac{2x}{\theta^2}$

$$L(\theta) = \prod_{i=1}^n \frac{2x_i}{\theta^2} = \underbrace{2^n \left(\prod x_i \right)}_{\text{Constant}} \theta^{-2n}$$

$$\hat{\theta}_{MLE} = X_n$$

25. $X_1, \dots, X_n \quad f(x) = \frac{1}{\theta} e^{-x/\theta}$

(a) $E[X] = \frac{1}{\theta} \Rightarrow \bar{X} = \frac{1}{\theta} \Rightarrow \hat{\theta}_{MoM} = \frac{1}{\bar{X}}$

$E[X] = \theta \Rightarrow \bar{X} = \theta \Rightarrow \hat{\theta}_{MoM} = \bar{X}$

(b) $L(\theta) = \theta^{-n} e^{-\sum x_i / \theta}$

$$l(\theta) = -n \log \theta - \frac{\sum x_i}{\theta}$$

$$l'(\theta) = -\frac{n}{\theta} + \frac{\sum x_i}{\theta^2} = 0$$

$$-n\theta + \sum x_i = 0$$

$$-n\theta = -\sum x_i \Rightarrow \theta = \frac{\sum x_i}{n} = \bar{X}$$

26. $X_1, \dots, X_n \sim \text{Bern}(p)$

(a) $E[X] = p$, $\bar{X} = p \Rightarrow \hat{p}_{\text{mom}} = \bar{X}$

(b) $P(X=x) = p^x (1-p)^{1-x}$

$$L(p) = \prod p^{x_i} (1-p)^{1-x_i} = p^{\sum x_i} (1-p)^{n-\sum x_i}$$

$$\ell(p) = (\sum x_i) \log p + (n - \sum x_i) \log(1-p)$$

$$\ell'(p) = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p} = 0$$

$$\frac{\sum x_i}{p} = \frac{n - \sum x_i}{1-p}$$

$$(1-p) \sum x_i = (n - \sum x_i) p$$

$$\sum x_i - p \sum x_i = np - p \sum x_i$$

$$\hat{p}_{\text{MLE}} = \frac{\sum x_i}{n} = \bar{X}$$

27. $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, σ^2 is known.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$L(\mu) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{n}{2\sigma^2} \sum (x_i - \mu)^2\right)$$

$$\sum (x_i - \mu)^2 = 0$$

$$\hat{\mu}_{\text{MLE}} = \bar{X}$$

28. $x_i \sim N(\mu, \sigma^2)$ $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$

$$L(\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2\right)$$

$$l = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$$

$$\frac{dl}{d\mu} = \frac{1}{\sigma^2} \sum (x_i - \mu) = 0 \Rightarrow \sum x_i - n\mu = 0$$

$$\hat{\mu}_{MLE} = \bar{X}$$

$$\frac{dl}{d\sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum (x_i - \bar{X})^2 = 0$$

$$n\sigma^2 = \sum (x_i - \bar{X})^2$$

$$\boxed{\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum (x_i - \bar{X})^2}$$

29. $X_1, \dots, X_n \sim \text{Poisson}$
 $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$

(a) $E[X] = \lambda$, $\bar{X} = \lambda \Rightarrow \hat{\lambda}_{MOM} = \bar{X}$

(b) $f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$

$$L(\lambda) = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod x_i!}$$

$$l(\lambda) = -n\lambda + \sum x_i \log \lambda - \sum \log(x_i!)$$

$$l'(\lambda) = -n + \frac{\sum x_i}{\lambda} \Rightarrow -n + \frac{\sum x_i}{\lambda} = 0$$

$$n = \frac{\sum x_i}{\lambda} \Rightarrow \lambda = \frac{\sum x_i}{n}$$

$$\hat{\lambda}_{MLE} = \bar{X}$$

30. $X_1, \dots, X_n$ i.i.d. $f(x) = \frac{3x^2}{\theta^3}$

(a) $E[X] = \frac{3\theta}{4}$, $\bar{X} = \frac{3\theta}{4} \Rightarrow \hat{\theta}_{MOM} = \frac{4\bar{X}}{3}$

(b) $f(x) = \frac{3x^2}{\theta^3}$

$L(\theta) = \prod_{i=1}^n \frac{3x_i^2}{\theta^3} = 3^n \left(\prod_{i=1}^n x_i^2 \right) \theta^{-3n}$

$\hat{\theta}_{MLE} = X_n$